



UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

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ADVANCED SIGNAL PROCESSING - *Prova scritta preliminare del 15/06/2016*

EXERCISE

We observe N samples of a realization of the random process $X[n] = A + W[n]$, for $n=0, 1, \dots, N-1$, where A a constant signal of interest (SOI), modelled as an unknown deterministic parameter, and $\{W[n]; n = 0, 1, \dots, N-1\}$ are independent and identically distributed (IID) random variables modeling additive noise, which is independent of A . Each noise sample is Gaussian distributed, zero mean and with unknown variance σ_w^2 . The signal to noise ratio (SNR) is as follows: $\gamma \triangleq A^2 / \sigma_w^2$.

- (1) Derive the autocorrelation function (ACF) $R_x[m] = E\{X[n]X[n+m]\}$ and the power spectral density (PSD) $S_x(e^{j2\pi f}) = FT\{R_x[m]\}$ of the observed signal $X[n]$ and plot them.
- (2) Derive the joint maximum likelihood (ML) estimator of the unknown signal A and unknown noise power σ_w^2 .
- (3) Derive the bias and the mean square error (MSE) of the ML estimator of A , plot the MSE as a function of γ and find the minimum value of N which guarantees an MSE lower than 10^{-3} when the noise power is $\sigma_w^2 = 1$.
- (4) Derive the bias of the ML estimator of σ_w^2 and plot it as a function of the sample size N .
- (5) Does it exist the linear Least Squares (LS) estimator of A ? Does it exist the LS estimator of σ_w^2 ?
[Motivate the answers, but it is not require to find these estimators, if they exist].



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Solution:

(1) Derive the autocorrelation function (ACF) $R_x[m] = E\{X[n]X[n+m]\}$ and the power spectral density (PSD) $S_x(e^{j2\pi f}) = FT\{R_x[m]\}$ of the observed signal $X[n]$ and plot them.

$$R_x[m] = E\{X[n]X[n+m]\} = A^2 + \sigma_w^2 \delta[m]$$

$$S_x(e^{j2\pi f}) = FT\{R_x[m]\} = 2\pi A^2 \delta(e^{j2\pi f} - 1) + \sigma_w^2$$

(2) Derive the joint maximum likelihood (ML) estimator of the unknown signal A and unknown noise power σ_w^2 .

$$\hat{A}_{ML} = \frac{1}{N} \sum_{n=0}^{N-1} X[n]$$

$$\hat{\sigma}_{w,ML}^2 = \frac{1}{N} \sum_{n=0}^{N-1} \left(X[n] - \hat{A}_{ML} \right)^2 = \frac{1}{N} \sum_{n=0}^{N-1} \left(X[n] - \frac{1}{N} \sum_{k=0}^{N-1} X[k] \right)^2$$

(3) Derive the bias and the mean square error (MSE) of the ML estimator of A , plot the MSE as a function of γ and find the minimum value of N which guarantees an MSE lower than 10^{-3} when the noise power is $\sigma_w^2 = 1$.

$$b\{\hat{A}_{ML}\} = E\{A - \hat{A}_{ML}\} = 0$$

$$MSE\{\hat{A}_{ML}\} = \text{var}\{\hat{A}_{ML}\} = \frac{\sigma_w^2}{N}$$

$$MSE\{\hat{A}_{ML}\} = \frac{\sigma_w^2}{N} < 10^{-3} \rightarrow N > 10^3 \sigma_w^2 = 10^3$$



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(4) Derive the bias of the ML estimator of σ_w^2 and plot it as a function of the sample size N .

$$\hat{\sigma}_{w,ML}^2 = \frac{1}{N} \sum_{n=0}^{N-1} (X[n] - \hat{A}_{ML})^2 = \frac{1}{N} \sum_{n=0}^{N-1} X^2[n] - (\hat{A}_{ML})^2$$

$$b\{\hat{\sigma}_{w,ML}^2\} = \sigma_w^2 - E\{\hat{\sigma}_{w,ML}^2\} = \sigma_w^2 - (A^2 + \sigma_w^2) + A^2 + \frac{\sigma_w^2}{N} = \frac{\sigma_w^2}{N}$$

(5) Does it exist the linear Least Squares (LS) estimator of A ?

Since the noise is AWGN, we have: $\hat{A}_{LS} = \hat{A}_{ML} = \frac{1}{N} \sum_{n=0}^{N-1} X[n]$

Does it exist the LS estimator of σ_w^2 ?

$\hat{\sigma}_{w,LS}^2$ does not exist, since the signal is not a linear function of the unknown parameter σ_w^2